

Galaxy Image Classification and the Galaxy Merger Rate

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We present a convolutional neural network (CNN) pipeline for morphological classification of galaxies from Sloan Digital Sky Survey (SDSS) images, using Galaxy Zoo 2 (GZ2) citizen-science labels as training data. Starting from 61,578 galaxy images with 37 continuous morphological labels (vote fractions in $[0, 1]$), we center-crop and resize images to 96×96 pixels ($\sim 20\times$ reduction in pixel count), split 80/20 into training (49,262) and validation (12,316) sets, and establish a mean-prediction baseline validation RMSE of 0.1639. We train two backbone architectures — a 4-block custom CNN (32/64/128/256 channels with dropout) and an 18-layer ResNet adapted for small inputs and sigmoid output — and apply two optimization techniques to each: **ReduceLROnPlateau** learning-rate scheduling and random $[0, 360^\circ)$ rotation augmentation exploiting the rotational invariance of galaxy morphology. The augmented ResNet with LR scheduling achieves the best validation RMSE of 0.0965, a $\sim 41\%$ improvement over the baseline. Per-label scatter plots reveal that top-level morphological features (smooth vs. disk) are predicted accurately, while rare sub-classifications (rings, mergers, lens/arcs) remain challenging due to sparse training examples. We apply the trained model to 79,975 test images and estimate a galaxy merger fraction of $\sim 4.3\%$, comparing to the $z \approx 0$ merger rate from J. M. Lotz et al. (2011) and discussing complicating factors including the distinction between merger rate (Gyr^{-1}) and instantaneous merger fraction.

1. Introduction

Understanding galaxy morphology is fundamental to tracing the evolutionary history of galaxies across cosmic time (C. J. Conselice 2014). The visual classification of galaxies into categories — elliptical, spiral, irregular, merging — dates back to Hubble’s original tuning fork diagram, and modern large-scale surveys have extended this classification to hundreds of thousands of galaxies through citizen science projects such as Galaxy Zoo (C. J. Lintott et al. 2008). Galaxy Zoo 2 (K. W. Willett et al. 2013) refined the classification scheme to 37 continuous labels organized in a hierarchical decision tree, enabling nuanced morphological descriptions including bar presence, spiral arm count, bulge prominence, and disturbed/merging features.

However, human classification does not scale to the billions of galaxies that will be observed by upcoming surveys (LSST, Euclid, Roman). Convolutional neural networks (CNNs) offer an automated alternative: given sufficient training data, a CNN can learn to extract morphological features directly from pixel data (A. Krizhevsky et al. 2017). In this work, we train CNNs on GZ2 labels to predict all 37 morphological classifications simultaneously, then apply the trained model to estimate the galaxy merger fraction at $z \approx 0$.

Figure 1 provides an overview of the analysis pipeline.

2. Data Exploration

The training set consists of 61,578 SDSS *gri*-composite images (424×424 pixels) and 37 GZ2 classification labels per galaxy. Each label is a vote fraction in $[0, 1]$ representing the proportion of human classi-

fiers who assigned that morphological feature. The labels are organized hierarchically: top-level questions (smooth vs. disk vs. star/artifact) gate whether deeper questions (bar? spiral? arm count?) are asked.

2.1. Label Distributions

Figure 2 shows normalized histograms of all 37 labels. Top-level labels (smooth, features/disk) exhibit broad, roughly bimodal distributions, while deeper labels (arm count, spiral tightness) are heavily concentrated near zero because most galaxies never reached those questions. This hierarchical concentration is a fundamental property of the GZ2 label space and presents a challenge for the CNN: the model must learn that most labels are near-zero for most galaxies.

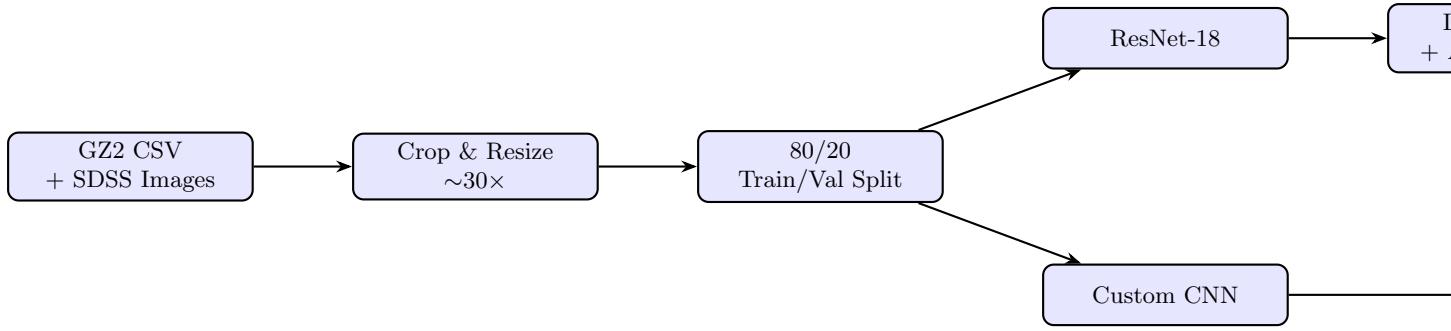


Fig. 1.— Analysis pipeline. SDSS galaxy images and GZ2 classification labels are preprocessed (crop, resize, split), then used to train a custom CNN and a ResNet-18. The best model is optimized with learning-rate scheduling and data augmentation, then evaluated on the validation and test sets. The final step estimates the galaxy merger fraction.

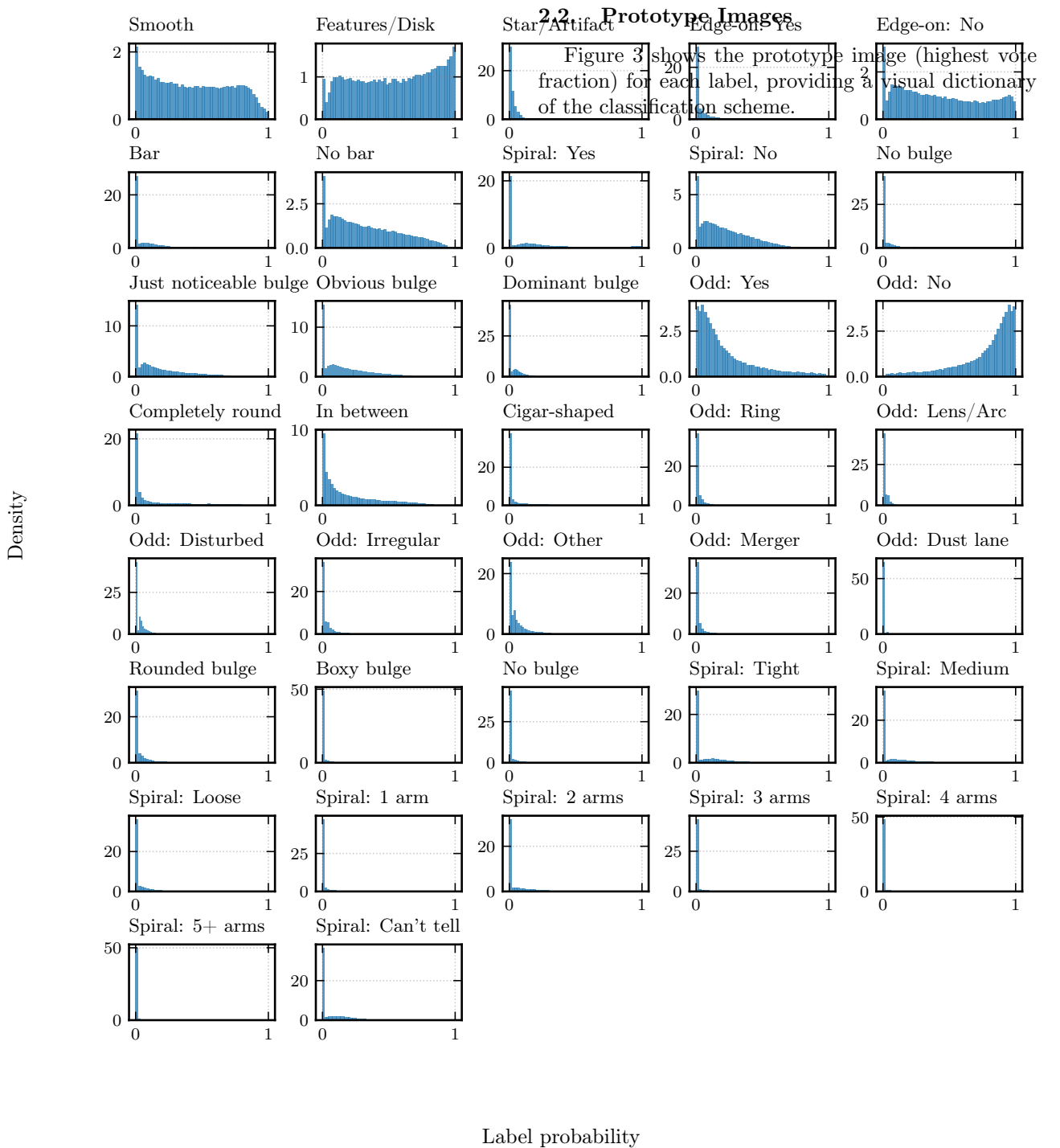
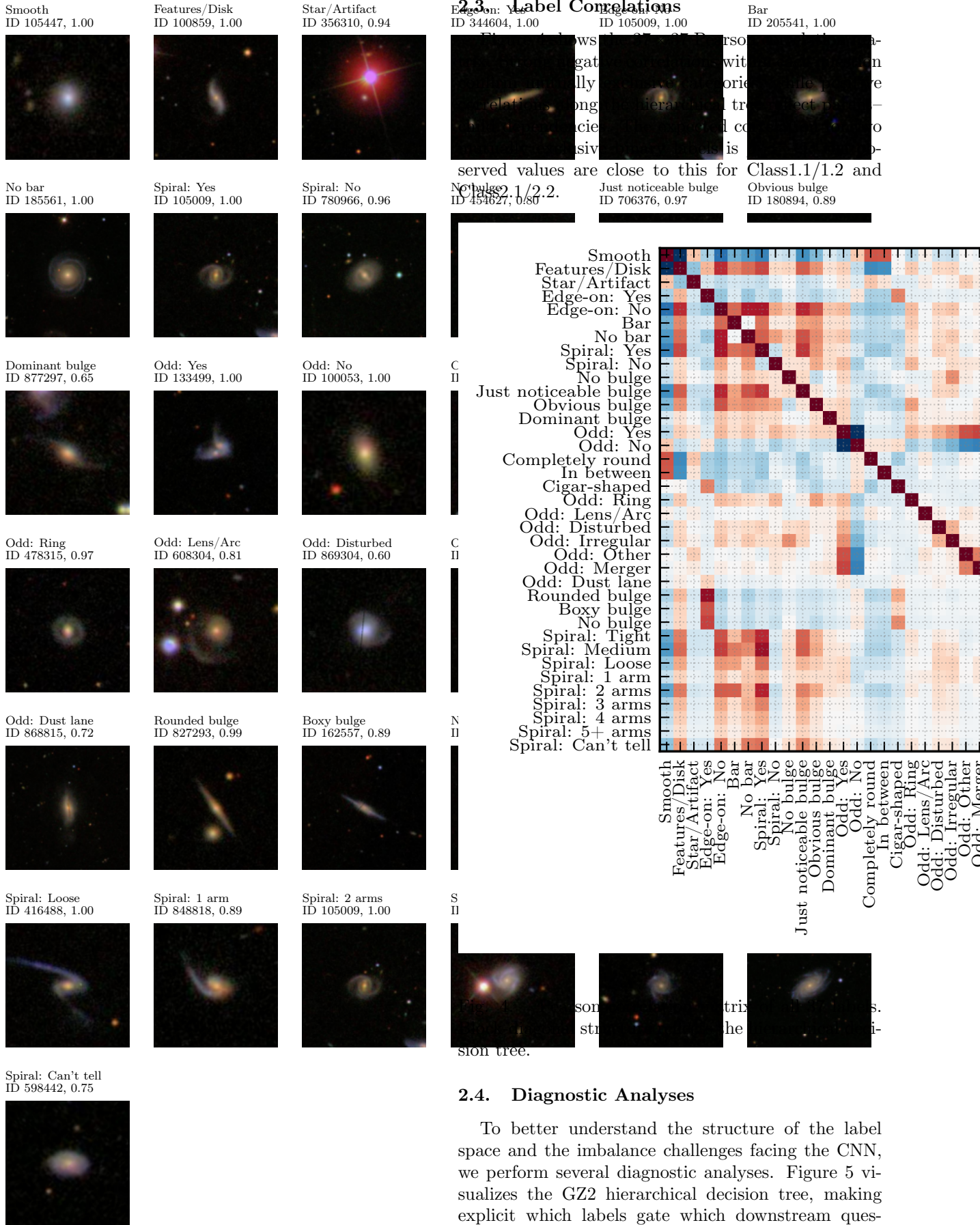


Fig. 2.— Normalized histograms of all 37 GZ2 classi-



2.3. Label Correlations

Heatmap showing correlations between 37 galaxy labels. The labels are listed on both the x and y axes. The color scale ranges from blue (low correlation) to red (high correlation). The diagonal is dark red, indicating perfect self-correlation. The heatmap shows a block-like structure, suggesting that labels within the same category (e.g., 'Smooth', 'Spiral') are highly correlated with each other.

Fig. 3.— Prototype images: the galaxy with the highest vote fraction for each of the 37 labels.

2.4. Diagnostic Analyses

To better understand the structure of the label space and the imbalance challenges facing the CNN, we perform several diagnostic analyses. Figure 5 visualizes the GZ2 hierarchical decision tree, making explicit which labels gate which downstream questions. Figure 6 reorders the correlation matrix by tree branch, exposing block structure that the unordered matrix obscures. Figure 7 shows a t-SNE embedding of the 37-dimensional label space; the manifold separates cleanly along the smooth/disk axis but rare

classes form sparse, isolated clusters. Figure 8 quantifies the effective sample size $N_{\text{eff}} = (\sum f_i)^2 / \sum f_i^2$ per label, demonstrating that several deeper-tree labels have $N_{\text{eff}} \lesssim 1000$, far below the nominal 61,578.

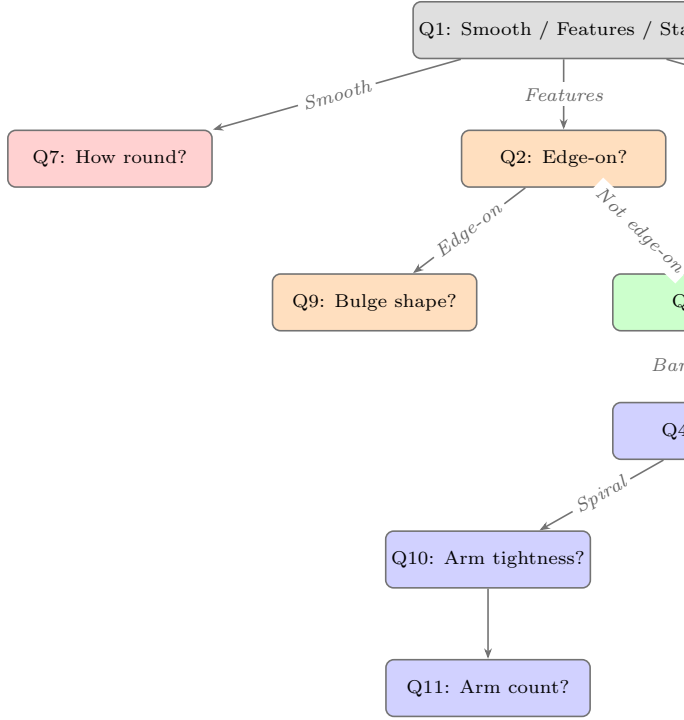


Fig. 5.— GZ2 hierarchical decision tree. Each node is a question; edges show which downstream questions are gated by each answer. Node colors denote branch: gray = root, red = smooth, orange = edge-on, green = face-on/bar, blue = spiral, violet = bulge, brown = odd-feature.

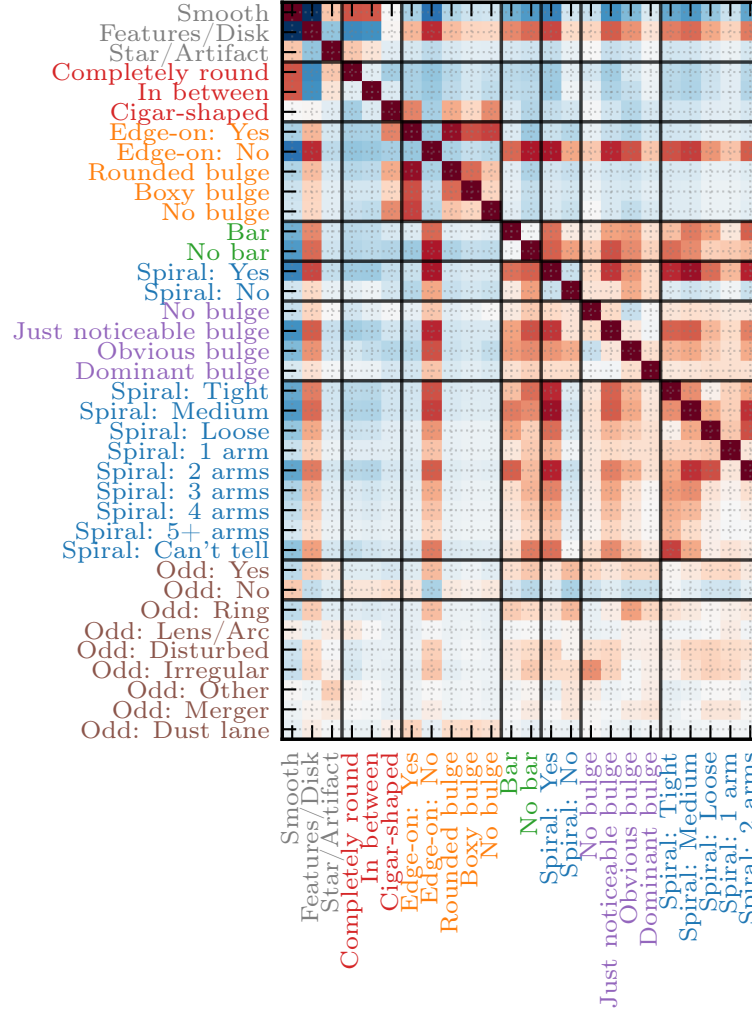


Fig. 6.— Correlation matrix reordered by tree branch, revealing block structure aligned with the hierarchy.

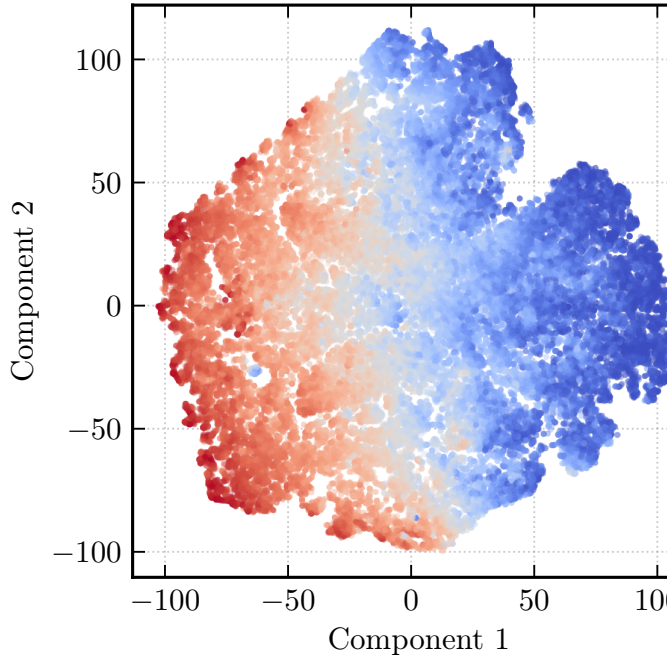


Fig. 7.— t-SNE embedding of the 37-dimensional label space, colored by dominant top-level class.

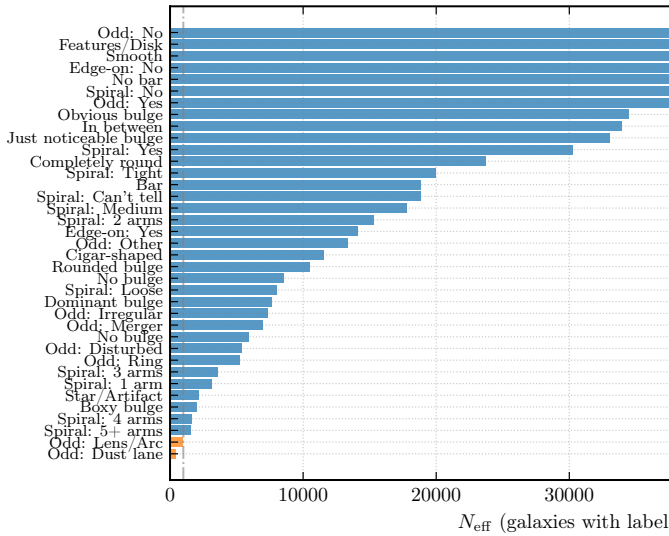


Fig. 8.— Effective sample size per label. Hierarchical gating drives deeper-tree labels to N_{eff} far below the nominal 61,578.

3. Data Preparation

3.1. Image Downsizing

Raw SDSS images are 424×424 pixels, of which most is empty sky. We center-crop (removing 25% of the border on each side) and resize to 96×96 pixels, achieving a $\sim 20\times$ reduction in pixel count while preserving morphological features (Figure 9).

Original — 495595



Original — 181525



Original — 687266



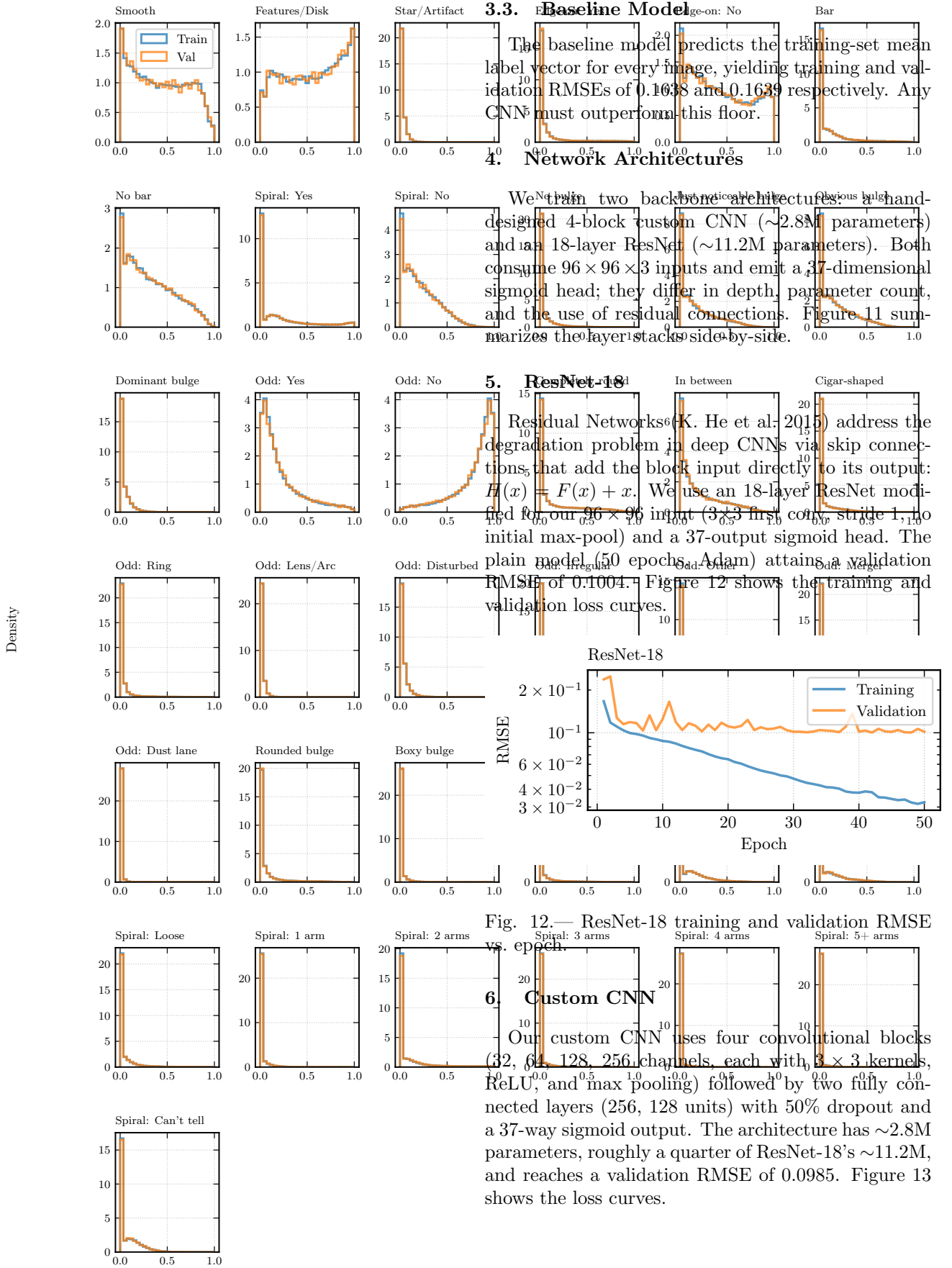
Original — 794293



Fig. 9.— Original vs. downsized images for four example galaxies.

3.2. Train/Validation Split

We randomly split the dataset 80/20 (seed 42) into 49,262 training and 12,316 validation images. Figure 10 confirms that the label distributions are statistically indistinguishable between the two sets.



Label probability

Fig. 10.— Training (blue) vs. validation (orange) label distributions.

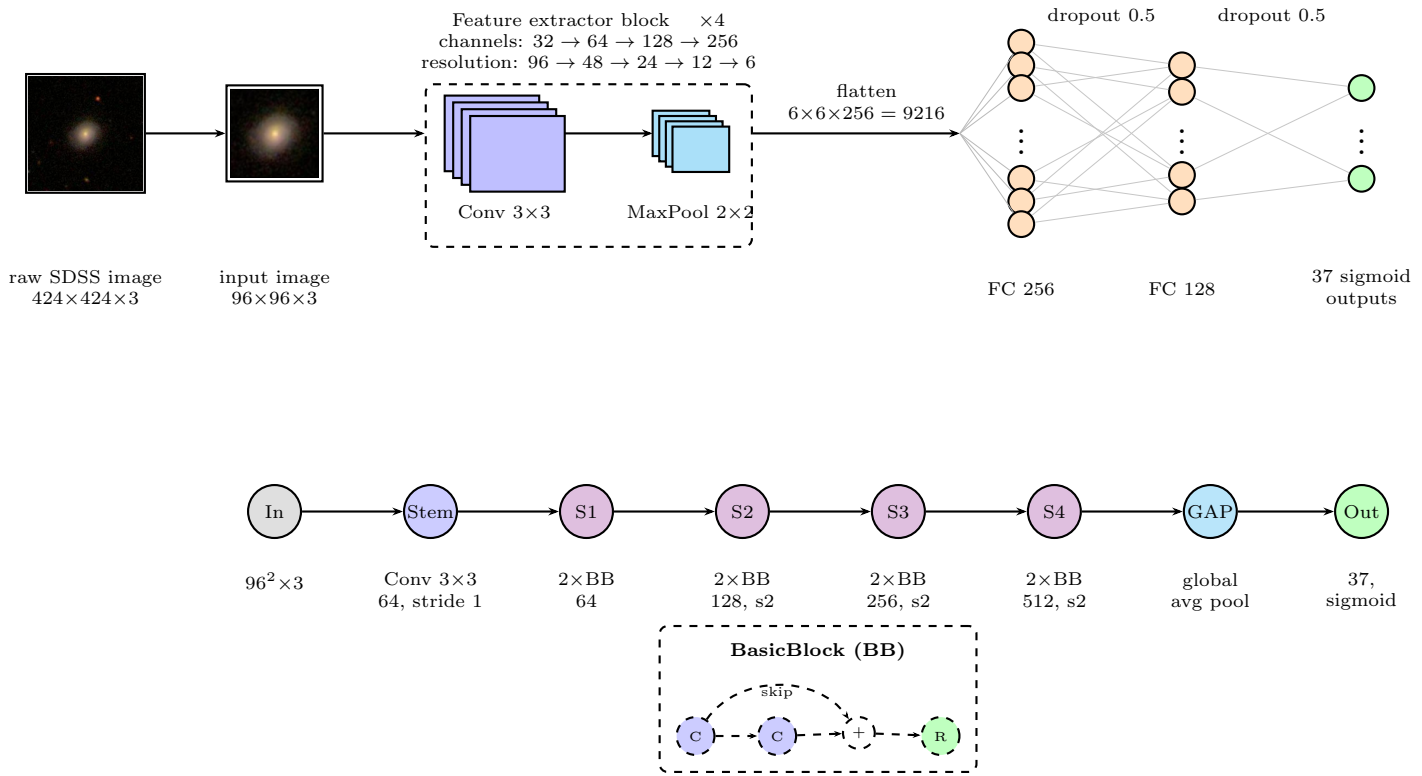


Fig. 11.— Architecture flow for the two backbones, read left-to-right. *Top* — *Custom CNN* ($\sim 2.8M$ params): a $96 \times 96 \times 3$ input image passes through a convolutional feature extractor consisting of a stack of 3×3 convolutions followed by 2×2 max-pooling, with the dashed loop indicating the block is repeated four times at channel widths $32 \rightarrow 64 \rightarrow 128 \rightarrow 256$. The flattened feature vector is then fully connected to 256 units, halved to 128 units, and finally projected onto 37 sigmoid outputs. Edges between FC layers are drawn schematically. *Bottom* — *ResNet-18* ($\sim 11.2M$ params): a 3×3 /stride-1 stem (no initial max-pool, suitable for the small 96×96 input) feeds four residual stages (S1–S4) of two BasicBlocks each at channel widths 64/128/256/512, ending in global average pooling and a 37-way sigmoid head. The dashed inset shows the internal flow of one BasicBlock: two 3×3 convolutions (C) summed with the identity skip (+) followed by ReLU (R).

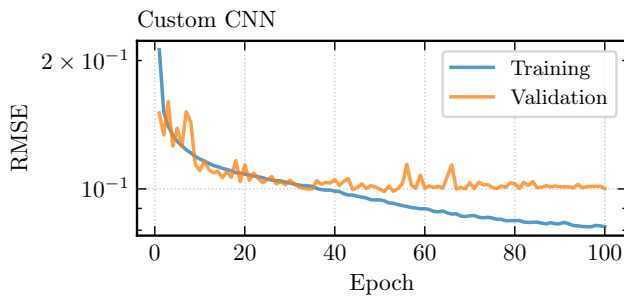


Fig. 13.— Custom CNN training and validation RMSE vs. epoch.

7. Optimization

7.1. Learning Rate Scheduling

We retrain both backbones with `ReduceLR0nPlateau` (factor 0.5, patience 3 epochs), which decreases the learning rate whenever the validation loss plateaus and enables finer convergence. Figures 14 and 15 show the resulting loss and learning-rate trajectories for the ResNet-18 and the custom CNN respectively. The

scheduler reduces the validation RMSE of the ResNet to 0.0977 and of the custom CNN to a comparable value.

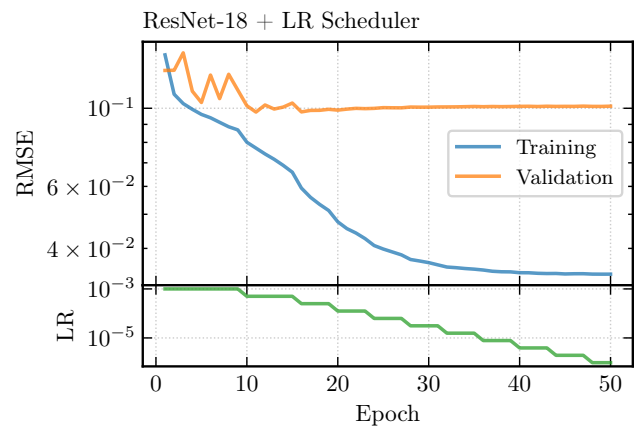


Fig. 14.— ResNet-18 with LR scheduler: loss curves (top) and learning rate (bottom).

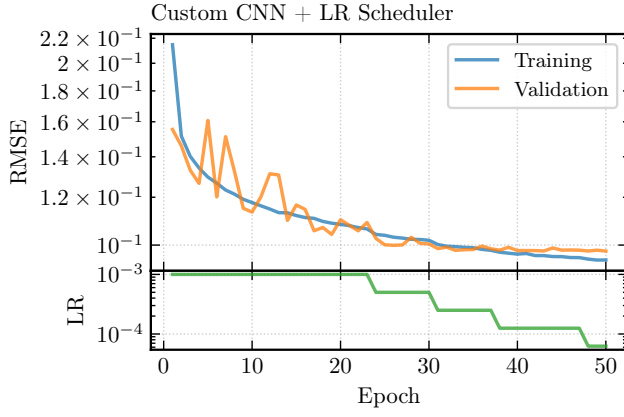


Fig. 15.— Custom CNN with LR scheduler: loss curves (top) and learning rate (bottom).

7.2. Data Augmentation

Galaxy morphological labels are rotationally invariant: a spiral galaxy remains a spiral under arbitrary rotation. We augment the training set by rotating each image by a uniformly random angle $\theta \in [0, 360)$ degrees, combined with the LR scheduler. Figures 16 and 17 show the improved loss curves with reduced overfitting for both backbones. The augmented ResNet attains the best overall validation RMSE of 0.0965.

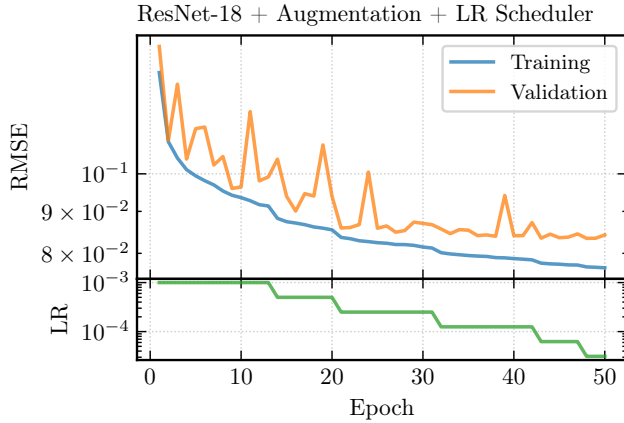


Fig. 16.— ResNet-18 with augmentation + LR scheduler.

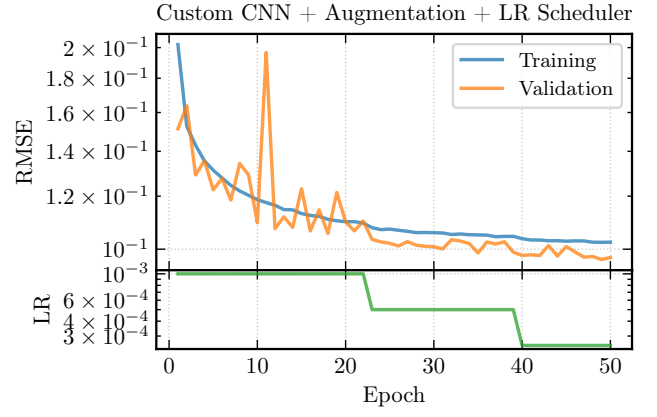


Fig. 17.— Custom CNN with augmentation + LR scheduler.

8. Model Comparison and Evaluation

8.1. Model Comparison

Figure 18 compares validation RMSE across all trained models. Table 1 summarizes the best validation RMSE for each. The ResNet-18 with augmentation and LR scheduling achieves the best performance, a $\sim 41\%$ reduction below the mean-prediction baseline.

Model	Best Val. RMSE
Mean-prediction baseline	0.1639
ResNet-18 (plain)	0.1004
Custom CNN (plain)	0.0985
ResNet-18 + LR scheduler	0.0977
ResNet-18 + augmentation + LR scheduler	0.0965

Table 1: Best validation RMSE achieved by each model.

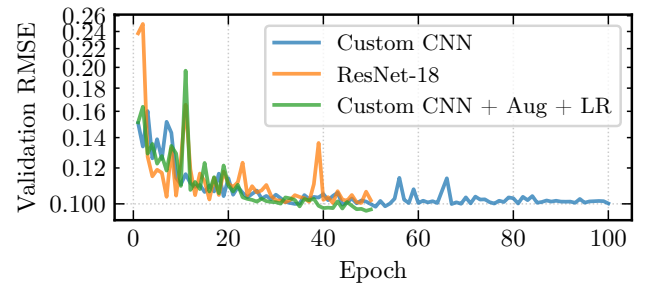


Fig. 18.— Validation RMSE comparison across three models.

8.2. Per-Label Performance

Figure 19 shows true vs. predicted scatter plots for all 37 labels on the validation set using the best model. Top-level features (smooth, disk) track the 1:1 line well; rare features (merger, ring, lens/arc) show larger scatter and regression toward the mean, reflecting the sparse training signal.



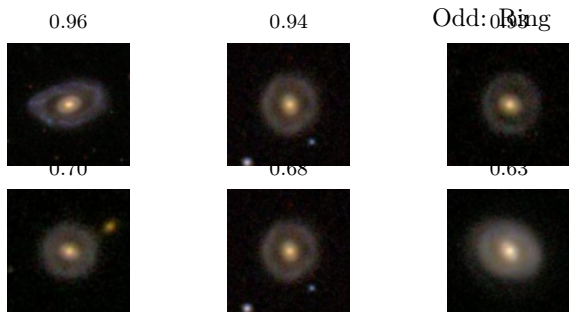


Fig. 23.— Top-5 images for “Odd: Ring.”

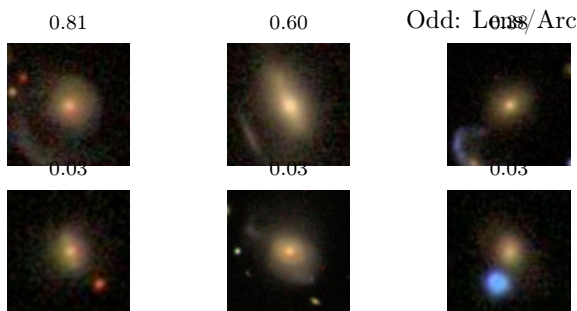


Fig. 24.— Top-5 images for “Odd: Lens/Arc.”

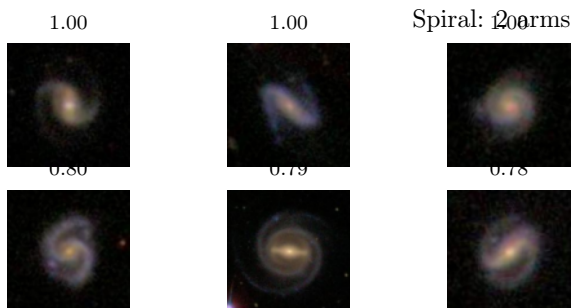


Fig. 25.— Top-5 images for “Spiral: 2 arms.”

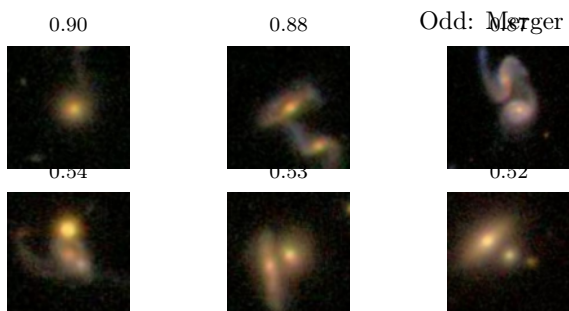


Fig. 26.— Top-5 images for “Odd: Merger.”

9. Merger Fraction

We apply the best performing model (augmented ResNet-18 with LR scheduling) to a held-out set of 1075 test images and estimate the galaxy merger fraction as the model-predicted probability of the “Odd: Merger” label (Class 86). We obtain a merger fraction of $\sim 4.3\%$. The Lotz et al. (2011) $z \approx 0$ major merger rate is $\sim 0.01\text{--}0.02 \text{ Gyr}^{-1}$; converting to an instantaneous merger fraction requires multiplying by the merger observability timescale $T_{\text{obs}} \sim 0.5\text{--}1 \text{ Gyr}$, giving an expected fraction of $\sim 0.5\text{--}3\%$, modestly below our estimate.

Several factors complicate direct comparison: (1) the GZ2 “merger” label conflates major and minor mergers, tidal interactions, and close pairs; (2) higher-redshift galaxies appear smaller and may be preferentially classified as disturbed; (3) the Lotz et al. rates are in units of Gyr^{-1} (a rate), while we estimate a dimensionless fraction; (4) GZ2 debiasing corrections do not fully account for resolution-dependent classification bias.

10. Conclusion

We have demonstrated that CNNs can predict Galaxy Zoo 2 morphological labels from SDSS images with RMSE well below the mean-prediction baseline. Data augmentation via random rotation and learning rate scheduling provide substantial improvements, exploiting the rotational invariance of galaxy morphology. The trained model enables automated estimation of the galaxy merger fraction on large image sets, although careful interpretation is required to compare to theoretical merger rates due to differences in definition and time scale.

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